Counterfactual conditionals: Processing dual representations

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It has been argued that counterfactual conditional statements elicit a dual representation in the mind of the of comprehender.[1] For instance, upon hearing the statement *If I had walked slower, I would have been hit by the car*, comprehenders envisage both a factual representation (I did not walk slowly and was not hit by the car) and a conjectural representation (I walked slowly and was hit by the car). Prior experimental and neuroimaging work found that both representations are active when processing counterfactuals.[2-4] Interestingly however, some of this research also finds that a subset of experimental participants never reaches a factual representation and only accesses the conjectural representation.[3] Here we explore whether this pattern is an experimental artifact or is due to true individual differences in counterfactual processing, by modifying the visual world paradigm that yielded these divergent results.

Thirty-seven adult native English-speakers were presented with a forced choice picture selection task, where they had to match indicative and counterfactual conditional statements with the appropriate picture. Unlike prior work that had participants inspect four pictures without committing to an interpretation, here we asked participants to explicitly choose between two pictures, corresponding to the factual and conjectural representations. The task was presented in the context of a story where Elmo and Big Bird bought fruits and vegetables for Cookie Monster. Cookie Monster then forgot what each friend bought and asked participants for help. Participants heard either an **indicative** (*if I find blackberries at the store, then I will buy them*) or a **counterfactual conditional** (*if I had found blackberries at the store, then I would have bought them*), and had to make a choice between two images (Table 1). One image depicted the mentioned fruit/vegetable (i.e., blackberries); the other an alternative fruit/vegetable (i.e., beans). Unlike prior research that represented the factual representation of the counterfactual as symbolic negation of the conjectural representation (i.e., an X on the mentioned fruit), we chose to represent negation through an alternative object, a method which is more natural than symbolic negation, and has also been used in past research.[4,5]

If participants compute the intended meaning of the conditionals, they should choose the image of the mentioned item in the **indicative**, but the image of the alternative (non-mentioned) item in the **counterfactual** (corresponding to the factual representation). If participants represent the dual meaning of counterfactuals, they should take longer to select the target picture in the counterfactual condition than the indicative condition.

A mixed effects analysis of the picture selection data showed that on average, participants were more likely to choose the target image when processing indicative rather than counterfactual conditionals ($M_{Indicative_target}$=0.98, $M_{Counterfactual_target}$=0.76, p<0.001) (Fig.1). A mixed effects analysis of reaction time showed that on average, participants took longer to respond for counterfactual than indicative conditionals ($M_{Indicative}$=1415ms, $M_{Counterfactual}$=2401ms, p<0.05) (Fig.1). To explore whether lower accuracy in the counterfactual condition was due to the performance of individual subjects, we looked more closely at each participant's performance and found 7 participants who consistently chose the non-target image for counterfactuals. Although this number is smaller than what was found in prior research, it indicates that despite experimental modifications, a subset of comprehenders consistently accesses only the conjectural representation of the counterfactual.

Overall, our study shows that, even when they are asked to commit to a single interpretation, comprehenders process counterfactual conditionals differently from indicative conditionals, with longer reaction times and greater inconsistency. Given the persistence of this pattern across different studies, an intriguing possibility is that these individual differences could be linked to lower performance in cognitive abilities linked to maintaining multiple representations in mind and interpreting speaker-intended meaning. Our on-going work is exploring this possibility.

References

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Table 1. Example of indicative and counterfactual conditional trials

Indicative conditional: *See these baskets? One has beans and the other has blackberries. One of these baskets is from Elmo. Elmo told me "If I find blackberries at the store, then I'll buy them." Which one is Elmo's basket?*	Target: blackberries
Counterfactual conditional: *See these baskets? One has beans and the other has blackberries. One of these baskets is from Big Bird. Big Bird told me "If I had found blackberries at the store, then I would have bought them" Which one is Big Bird's basket?*	Target: beans
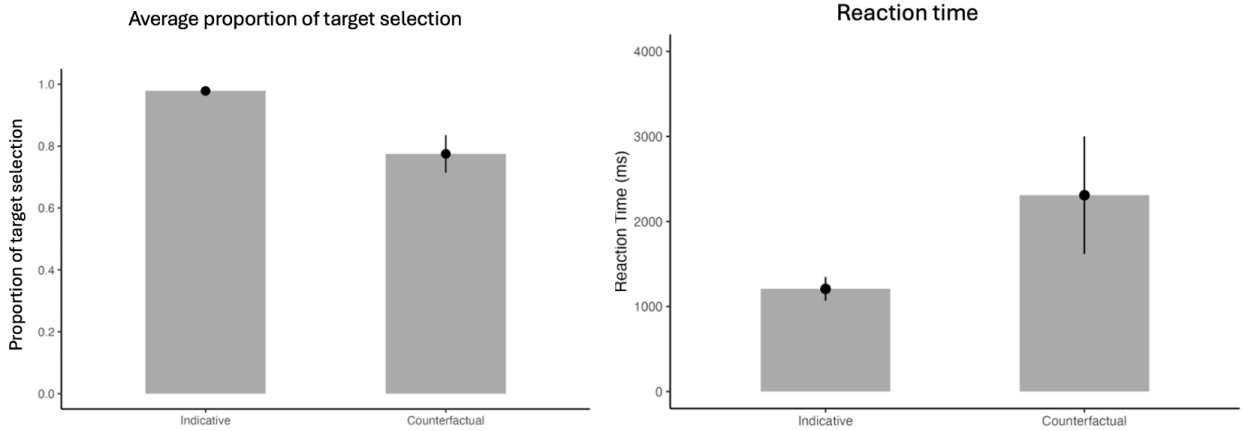
Figure 1. Average proportion of target selection (left) and reaction time for indicative and counterfactual conditionals (right). Error bars indicate standard error of the mean.